Open WebUI AI Security Lab Guide

A practical checklist of experiments to run against a local Open WebUI + llama.cpp setup for learning AI security concepts such as prompt injection, memory poisoning, context manipulation, and RAG attacks.

1. Baseline Prompt Injection

• Ask the model to reveal hidden instructions.

• Try variants such as 'Ignore previous instructions' and roleplay-based attacks.

• Compare success and failure cases.

2. System Prompt Extraction Attempts

• Create a custom system prompt.

• Ask the model to repeat or summarize its hidden instructions.

• Test direct, indirect, and translation-based approaches.

3. Document-Based Prompt Injection

• Upload a document containing instruction-like text.

• Ask the model to summarize the document.

• Observe whether document text is treated as data or authority.

4. RAG Poisoning

• Upload conflicting documents.

• Example: one says vacation policy is 14 days, another says 365 days.

• Observe retrieval ranking and answer quality.

5. Memory Poisoning

• Enable memory features.

• Ask the assistant to remember unusual instructions.

• Check what persists across conversations.

6. Context Flooding

• Provide large amounts of irrelevant text.

• Observe whether important instructions are displaced.

• Compare behavior with short and long contexts.

7. Role Confusion Tests

• Upload conversations that contain fake User/Assistant/System messages.
• Check whether the model incorrectly treats examples as instructions.

8. Contradictory Instruction Testing

• Provide multiple competing instructions.
• Measure which instruction source the model prioritizes.

9. Model Comparison

• Run the same attacks against different models.
• Compare robustness and consistency.

10. Logging and Observation

• Inspect prompts, retrieved chunks, memory entries, and model outputs.
• Focus on trust boundaries rather than only successful attacks.

Suggested Learning Progression

Open WebUI → Document Uploads → Prompt Injection → RAG Poisoning → Memory Poisoning → Aider → Flowise/Open Interpreter → OpenHands